QUALYXUS

Air-Gapped Edge AI for Visual Inspection & Process-Quality Intelligence

- **Startup Name:** Qualyxus
- **Founder & Lead Architect:** Utkarsh Sengar (B.Tech CSE, IIT Dharwad)
- **Prepared for:** 1517 Fund · Medici Project · November 2026

> **We solve** late-stage scrap and escaped defects **by providing** an on-premise edge-AI platform that inspects every part and links each defect to the machine cycle that produced it, **to help** automotive, aerospace & defense, and industrial manufacturers **catch** process drift and machine wear before they become scrap or downtime, with all data staying inside the plant.

Qualyxus at a Glance

The Whole Pitch on One Page

Problem

Cameras say a part is bad but not why; machine data never sees which parts failed. Scrap is found late and its cause stays hidden.

Solution

On-premise edge AI that inspects every part and links each defect to the machine cycle that made it, flagging process drift and machine wear before they cause scrap.

Built Today

Working platform: AUROC 0.95–0.99 on the public VisA benchmark, 1.8% calibrated false rejects, Modbus TCP / OPC UA cycle linking, advisory PLC output, 75 automated tests.

Market

$590M serviceable market: ~170,000 inspection points across India and SE Asia. First focus: CNC machining and injection moulding in Indian auto-component clusters.

Business Model

~$2.5k setup + ~$230 / month per line in India. Projected customer value $7–17k per line per year, payback in 4–10 months, 80%+ software margin.

What a Medici Grant Funds

A bench test rig: an NVIDIA Jetson edge computer, an industrial camera and lighting, so the platform runs on real hardware and real parts for the first time.

Stage: working prototype, pre-revenue; results so far are from public benchmarks and simulation. **Next:** real-hardware benchmarks on the test rig, then a first pilot in a nearby plant.

Compelling Problem Statement

The Problem: Quality and Machine Data Live in Separate Silos

Plants run two disconnected systems, so the defect and its cause are rarely seen together:

1. Machine Vision (Post-Mortem Sorting)

- Inspects parts **after** they are made, so scrap is caught only after material, machine hours and labour are spent.
- Rule-based setups need re-programming for every new part and struggle with variable cosmetic defects.
- Says pass or fail, but not **why**: blind to the machine conditions that caused the defect.

2. Machine Telemetry (Blind to Outcomes)

- Monitors temperatures, pressures, spindle load and vibration to prevent breakdowns.
- Never learns which parts came out bad, so it cannot tell which signal patterns cause defects.
- A machine can stay inside every alarm limit while slowly drifting into producing scrap.

Target Customer Profile: Plant Heads, VPs of Quality and Maintenance Heads running automated machining, moulding, casting, stamping, welding and assembly lines in automotive, aerospace & defense, and industrial manufacturing.

Solution Offered & Innovation Proposed

The Causal Bridge: Every Defect Linked to Its Machine Cycle

Qualyxus fuses camera inspection with machine-cycle data on one on-premise edge box, so quality and maintenance teams see the same cause.

Camera + Machine Data
Industrial cameras; Modbus TCP / OPC UA

On-Prem Edge Engine
Inspect, link to cycle, score quality & wear

Advisory PLC Signal
Reject / alert flag; plant logic decides

Core Deep-Tech Innovations

1 Learns from Good Parts

Anomaly detection (Dinomaly / PatchCore on DINOv2) trains on a few hundred good parts, with no defect library needed. AUROC 0.95–0.99 on 4 categories of the public VisA benchmark.

2 Calibrated False Rejects

Reject thresholds are set by statistical (conformal) calibration with a stated confidence bound, so the plant knows its false-reject rate before go-live.

3 Root Cause & Machine Health

Each part is linked to the cycle that made it. Cycle drift flags process problems and component wear (check rings, tools, dies) before they cause scrap, with approval-gated fixes (simulation-validated).

4 Air-Gapped by Design

Runs fully on-premise on an NVIDIA GPU; no cloud needed in production. Shadow mode, one-command rollback and a tamper-evident audit log built in.

Where We Play

One Platform Across Automated Manufacturing Processes

The vision model learns any part from good samples; the cycle engine reads any machine over Modbus TCP / OPC UA, and the same signals track the wear behind the defects.

Process	Defects the camera sees	Machine signals that explain them	Wear it tracks	Status
Injection Moulding	Short shots, flash, sink marks, burns, warpage	Injection & cavity pressure, melt / mould temperature, cushion	Check ring, heater bands, mould cooling	**Built end-to-end; validated in simulation**
CNC Machining	Burrs, chatter and tool marks, missing features	Spindle load, accelerometer vibration, tool counter	Cutting tools, spindle bearings	**Vision + OPC UA ready; MTConnect / FOCAS next**
Die Casting & Forging	Surface porosity, cold shuts, cracks, flash	Shot speed, die temperature, intensification pressure, press force	Dies, shot-cylinder seals	**Vision + cycle engine ready; template next**
Stamping & Press Work	Cracks, splits, wrinkles, burrs	Tonnage curve per stroke	Dies and punches	**Vision + cycle engine ready; template next**
Welding	Spatter, surface porosity, undercut, missing welds	Current, voltage, wire-feed speed	Electrode tips, wire feeder	**Vision + cycle engine ready; template next**
Assembly & Final Inspection	Missing, misaligned or wrong-variant parts	Press-fit force, torque curves	Press and nutrunner tooling	**Vision ready; multi-camera verdicts built**

Product & Service Details

Inspection Station, Safe PLC Integration & Model Operations

Inspection Station

- Industrial camera(s), lighting and an NVIDIA edge computer mounted at the line.
- **Multi-camera part verdicts:** a part passes only when every required view is inspected.
- Defect segmentation and typing (RF-DETR) once labelled examples exist.
- **"All-Faces" 360° enclosure:** in development.

Advisory PLC Integration

- **Advisory only:** writes a reject / alert flag to a standard PLC holding register over Modbus TCP.
- The plant's own PLC logic decides whether to divert or stop; certified safety functions are never modified.
- Designed to stay outside the safety-related control system (ISO 13849), so existing safety validation stays intact. Confirmed per site with the plant's safety engineer.

On-Prem Dashboard & Model Ops

- Live scrap rate, defect types, cycle risk, machine-health trends and part-position drift.
- Shadow mode, one-command rollback, drift alerts and approval-gated retraining.
- **Tamper-evident audit log** of every verdict: evidence for IATF 16949 / AS9100 audits.
- Review queue with labelling export (Label Studio / COCO).

Built & Measured

What the Platform Already Does

0.95–0.99

Anomaly-detection AUROC on 4 categories of the public VisA benchmark (previous baseline: 0.56–0.96).

1.8%

False-reject rate on held-out good parts with conformal-calibrated thresholds (synthetic parts).

100% → 6.7%

False rejects on loosely fixtured parts after automatic part registration, with defect recall unchanged (synthetic).

0.83

Mask IoU for defect segmentation (RF-DETR-Seg): 100% recall and defect-type accuracy, 16.5 ms per image on a laptop GPU (synthetic).

~4 s

To promote a new model live with no downtime; 98.5% shadow-mode agreement in closed-loop simulation.

129 / 129

Audit records verified intact in the hash-chained, tamper-evident log; 75 automated tests in passing.

Results on the public VisA benchmark and on our synthetic part and process simulator. Measuring the same metrics on real lines is the goal of the paid pilots.

Product Roadmap

Built Today → Next 6 Months → 12–18 Months

Built Today

- Good-parts-only anomaly detection with calibrated reject thresholds
- Defect segmentation & typing; part registration and tiling
- Multi-camera part verdicts; review & labelling loop
- Cycle engine (Modbus TCP, OPC UA), part-to-cycle linking, risk model and approval-gated fixes; MQTT / UNS publishing
- Model registry (shadow, rollback), tamper-evident audit log, on-prem dashboard, advisory PLC output

Next 6 Months (pilot phase)

- TensorRT deployment and benchmarks on NVIDIA Jetson Orin
- CNC connectors: MTConnect and FANUC FOCAS (spindle load, tool data)
- Accelerometer input with ISO 10816 vibration zones
- Offline installer with bundled models
- Supervised risk models trained on pilot data
- Machine-health index per component + maintenance log

12–18 Months

- "All-Faces" 360° multi-camera enclosure
- Process templates: die casting, stamping, welding
- Encrypted cross-plant federated learning (built, off by default)
- IATF 16949 / AS9100 audit-report packs
- Remaining-life estimates and maintenance planning (CMMS)
- MES / ERP integrations

Market Research & Opportunity

TAM / SAM / SOM — Bottom-Up Sizing

$45B

TAM (Total Addressable Market)

~₹3.9 lakh crore — Global AI visual inspection (~$31B) + predictive maintenance (~$14B), 2025 estimates.[1]

$590M

SAM (Serviceable Addressable Market)

~₹5,100 Cr / yr — ~1.7 lakh inspection points on machining, moulding, casting & forging, stamping, welding and assembly lines (~1.3 lakh India, ~0.4 lakh SE Asia) × ~₹3L blended contract.[2]

$5.5M

SOM (Serviceable Obtainable Market)

~₹48 Cr ARR — 2,000 lines in Indian industrial corridors (Pune, Chennai, Bengaluru, NCR, Hyderabad, Hubballi–Belagavi) × ₹2.4L list / yr. 5-year target (~1.5% of India); exports additive.

Market Tailwinds & Growth Drivers

Auto-Component Boom

₹7.6 lakh crore turnover in FY26, up 12.7% year on year, as global OEMs shift sourcing to India.[3]

India PLI Expansion

₹1.97 lakh crore PLI outlay across 14 sectors, incl. auto components, ACC batteries and electronics.[4]

Large Installed Base

₹31,781 Cr of machine tools consumed in FY25[5], ~5,000 foundries and 500+ forging units[6]: retrofit beats ₹50L+ replacement.

[1] MarketsandMarkets, Market Research Future; IMARC. [2] Qualyxus estimate (see notes). [3] ACMA, Jul 2026. [4] PIB, Nov 2024. [5] IMTMA (provisional). [6] Foundry Informatics Centre; AIFI. ₹87 = $1.

Competition & Comparison Matrix

Where Qualyxus Fits

	Legacy Vision (Cognex / Keyence)	AI Vision Competitors (SwitchOn)	Heavy Automation (Siemens / Bosch)	QUALYXUS PLATFORM
Operational Method	Post-Mortem Sorter (catches scrap late)	Primarily Visual Inspection	Heavy Physical Sensor Wiring	**Inspection + Early Warning on Process Drift & Wear**
Data Underpinnings	Rule-Based Image Tools (+ add-on deep learning)	Deep-Learning Image Inspection	Machine Telemetry Logs	**Images Linked to Machine-Cycle Data**
Deployment Time	Weeks of Integrator Programming per Part	Vendor-Led Setup per Part	Multi-Month Projects	**1–2 Day Non-Invasive Install (No Machine Rewiring)**
Safety Compliance	N/A (Post-Line)	N/A (Visual Check)	Changes May Need Safety Re-validation	**Advisory Only (No Safety-Function Changes)**
Budget Captured	Quality Control Only	Primarily Quality Control	Maintenance Only	**Dual Budget: Quality + Maintenance**
Typical Cost to Plant (indicative)	₹1.2L – ₹7.5L per smart camera + integration (one-time)	Subscription (pricing not public)	₹50L+ CapEx projects	**₹2–2.5L setup + ₹20k / month per line**

Cognex In-Sight 2000 to 8000-series dealer listings in India (ex-tax, lens and integration extra). Ranges are indicative and vary by application.

Business Model & Unit Economics

Hybrid Hardware-Enabled SaaS

Two revenue streams per line: a one-time hardware fee priced near cost-plus, and a recurring software licence that carries the margin.

1. One-Time Setup & Hardware: ₹2.0L – ₹2.5L per line

- Single-camera station: NVIDIA Jetson Orin-class edge computer, industrial camera + lens, lighting, enclosure, 1–2 day install.
- Multi-camera "All-Faces" station: ₹5L – ₹7L.
- Usually within plant-level purchase authority — no board-level CapEx cycle.
- Target hardware gross margin: **20–30%**

2. Software Licence: ₹20,000 / month per line

- ₹2.4L / year, billed annually: models, retraining, updates, dashboard, on-site support.
- Costs no more than one inspector (₹20–35k / month loaded), yet covers all three shifts.
- Volume: ₹2.0L / line / yr for 10+ lines. Extra camera: ₹5,000 / month. US / EU: $10k – $15k / line / yr.
- Machine-health add-on (from pilot phase): from ₹5,000 / machine / month. Target gross margin: **80%+**

Customer Payback (per line)

Metric	Value
Annual value delivered	₹6L – ₹15L typical; up to ₹25L on high-value parts
Year-1 cost to customer	₹4.4L – ₹4.9L (hardware + licence)
Payback period	4 – 10 months
Year-2+ return on licence	2.5× – 6×

Qualyxus Unit Economics (targets)

Metric	Target
Lines per plant (within 24 months)	5 – 8
Customer acquisition cost per plant	₹5L – ₹8L (pilot + field sales)
Gross profit per plant at 6 lines	~₹11.5L / year
CAC payback	5 – 9 months

All figures are planning assumptions to be validated in paid pilots.

Team & Advisory Details

Leadership, Cap Table & Strategic Advisors

Founder & Core Architect

Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

- Built the full platform: good-parts-only anomaly detection, defect segmentation, part-to-cycle linkage and risk models, edge deployment and model operations.
- LinkedIn: linkedin.com/in/utkarsh-sengar

Cap Table Status

100% Founder Owned (Utkarsh Sengar)

Company incorporation planned after the first pilot.

Strategic Advisory Board

Dr. Rakesh Kumar

VP R&D, Rossell Techsys (ex-Moog, ex-Indian Navy).

Shankar Lal

Discipline Chief, Cyient (21 years at HAL).

Ramesh Rana

Operations Lead, SIG SAUER.

What the Medici Grant Would Fund

Taking Qualyxus from Benchmarks to Real Cameras on Real Parts

The Gap Today

- Everything so far runs on the public VisA benchmark and my own process simulator.
- Factories will only trust numbers measured on real hardware, with real lighting and real parts.
- A small test rig closes that gap and becomes the station I carry into the first plant pilot.

Bench Test Rig (~$1,000)

Item	Approx. cost
NVIDIA Jetson Orin Nano developer kit (edge computer)	~$400
Industrial camera + lens	~$400
LED lighting, mounts and sample parts	~$200
Total	~$1,000

What It Proves (within ~8 weeks)

- On-device speed and accuracy on real parts, published openly.
- A working station ready for the first pilot in a nearby plant.

Costs are Indian retail estimates converted at ₹87 = $1.

QUALYXUS

Edge AI that traces every defect to its cause

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Demo
Watch the demo video